Introduction to Cloud Computing

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Seminar for: BS(CS) 1st semester

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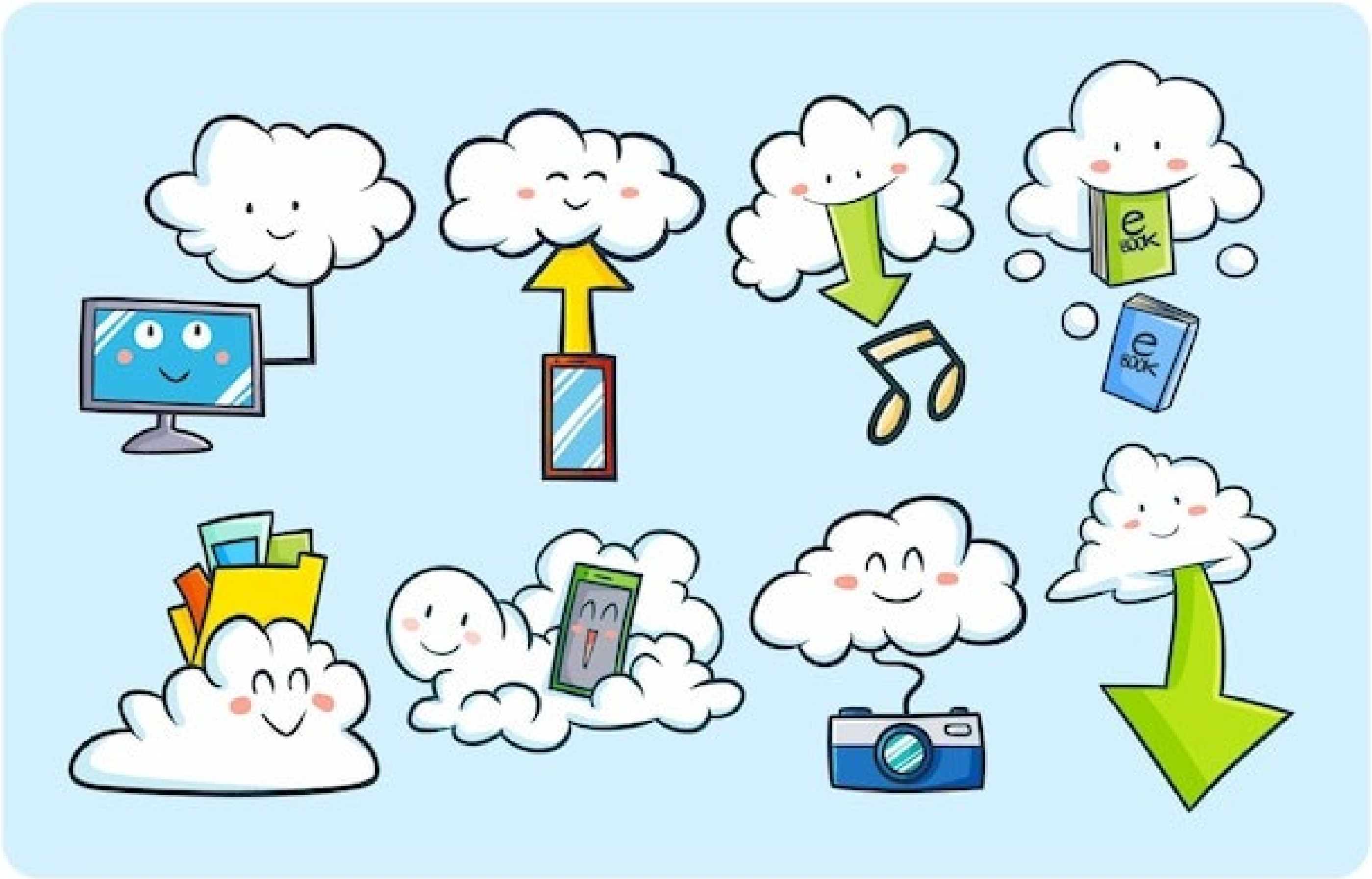
Agenda

- Introduction to Cloud Computing
- Core Components of Cloud Computing
- Cloud Services Models
- Cloud Deployment Models
- Major Cloud Services Providers
- Benefits of Cloud Computing
- Future Trends in Cloud Computing
- Q & A Session

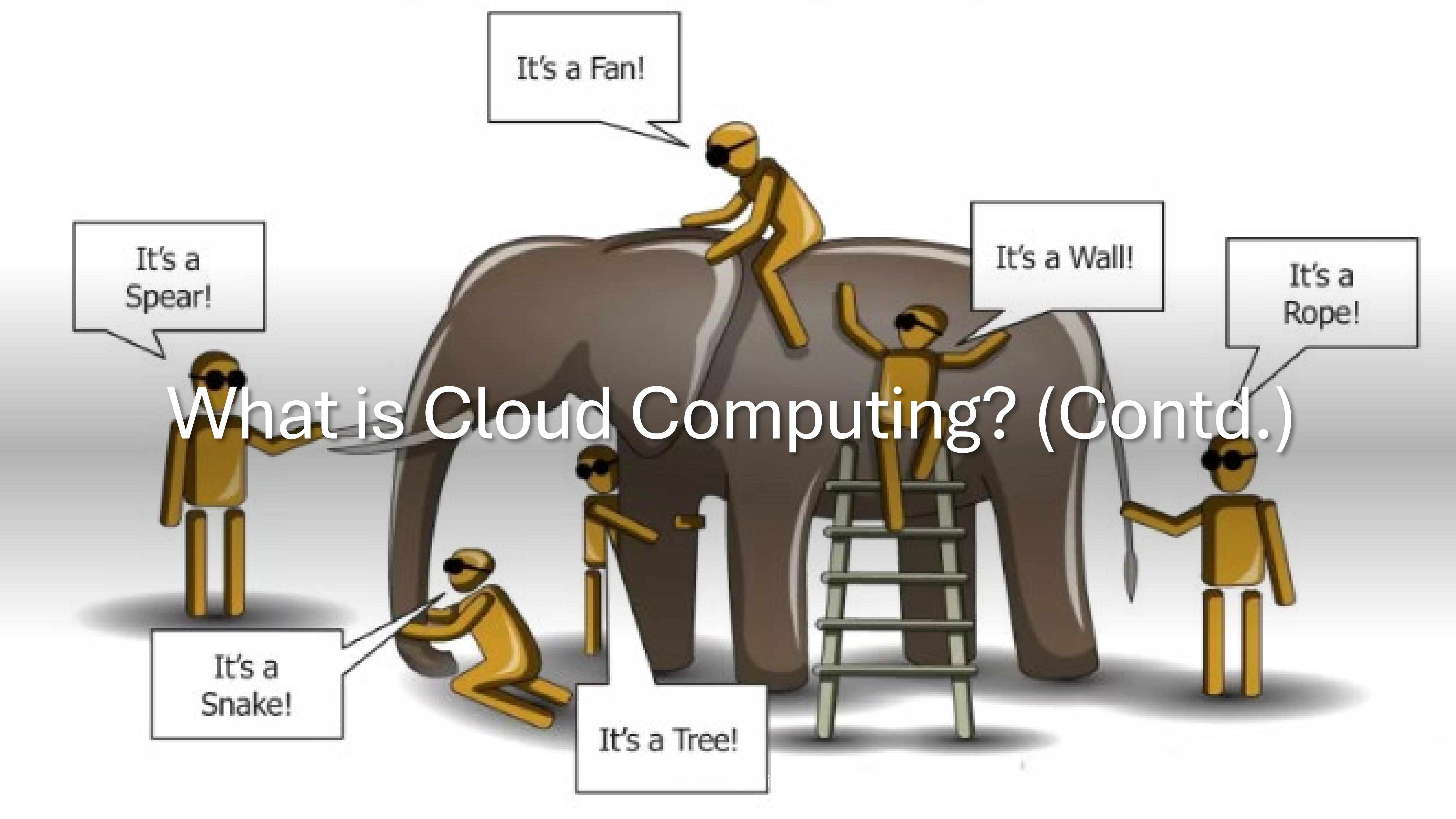
Kashif Ali

- Inside Solutions Architect at Zones, LLC
- 12+ years IT industry experience
- Lecturer at The University of Agriculture, Peshawar and UET Peshawar
- Trainer of Virtualization and Cloud Technologies at Corelinks Peshawar
- AWS Certified Solutions Architect – Associate
- Microsoft Certified: Azure Administrator Associate
- VMware vExpert 2021-24
- 42+ IT Certifications

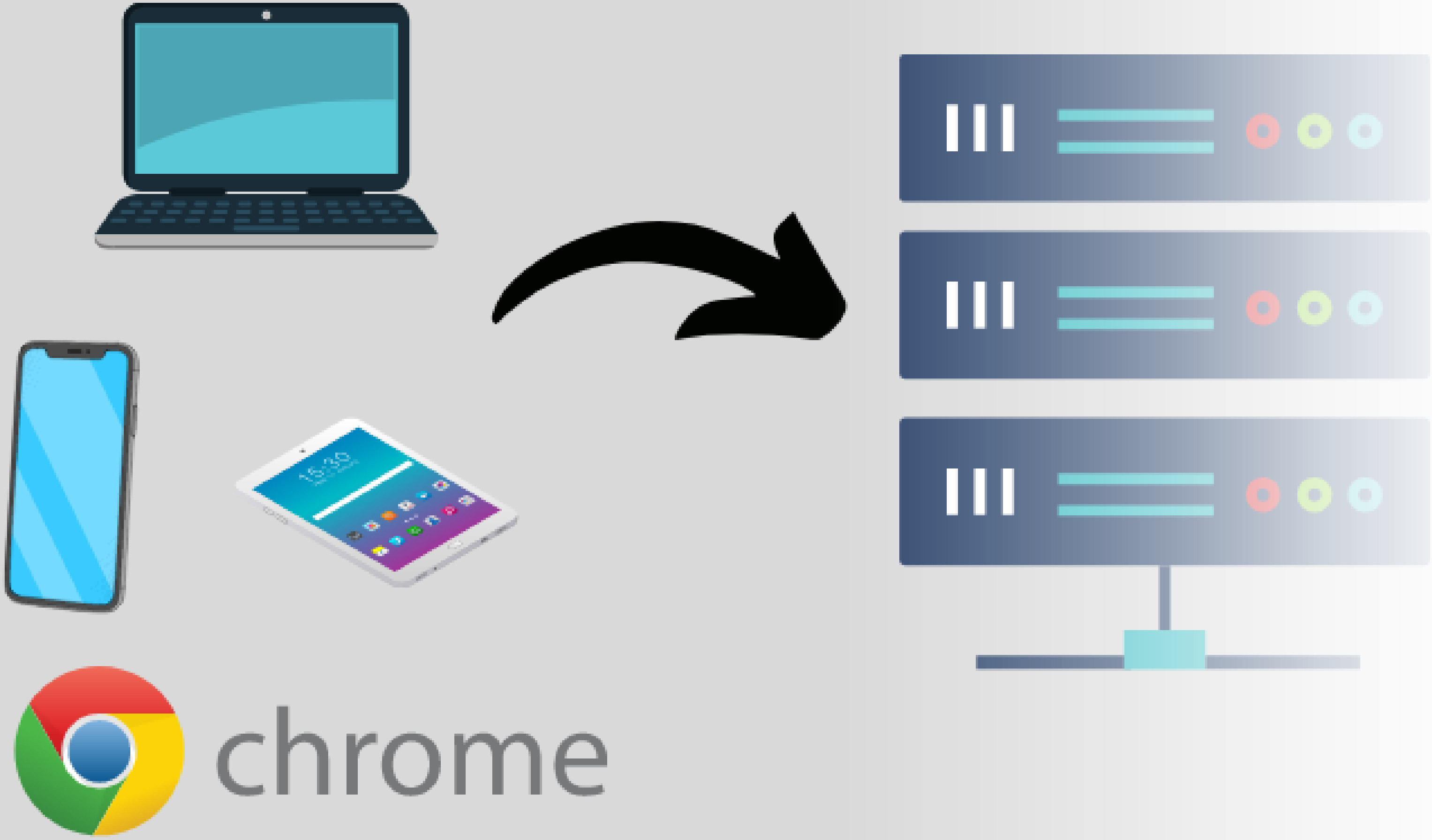
What is Cloud Computing?



What is Cloud Computing? (Contd.)



Client   Server



What is a client-server model?

In computing, a client can be a web browser or desktop application that a person interacts with to make requests to computer servers.

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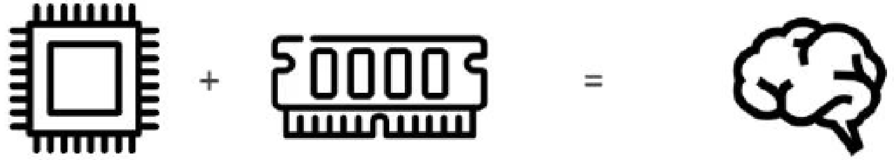
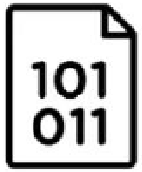
What is a server composed of?

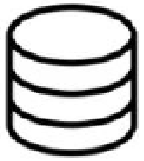
- Compute: CPU
- Memory: RAM



- Storage: Data

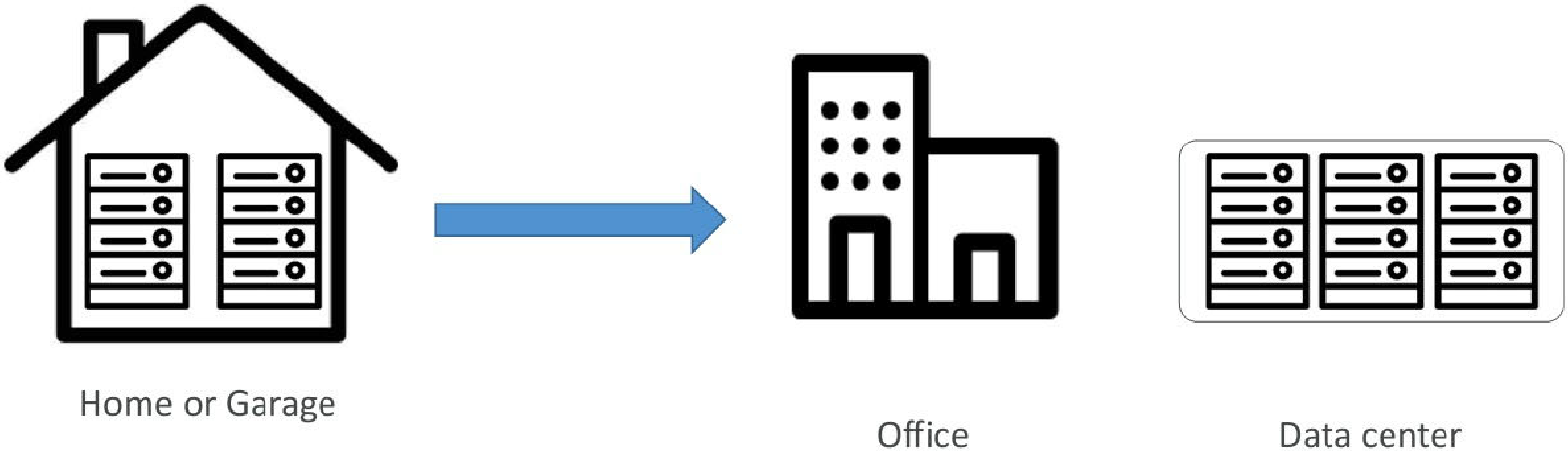


- Database: Store data in a structured way



- Network: Routers, switch, DNS server

Traditionally, how to build infrastructure



Problems with traditional IT approach

Pay for the rent for the data center

Pay for power supply, cooling, maintenance

Adding and replacing hardware takes time

Scaling is limited

Hire 24/7 team to monitor the infrastructure

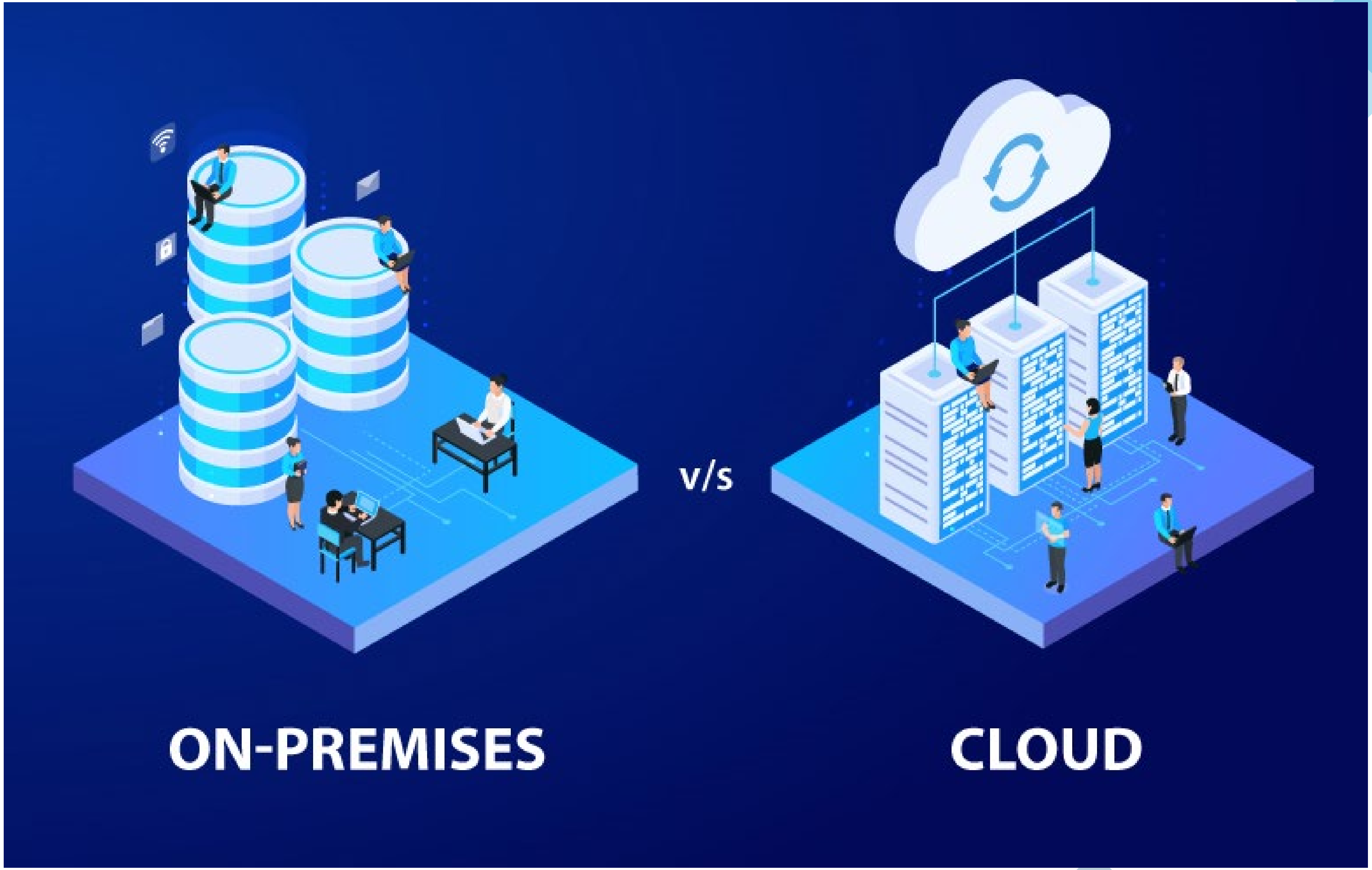
How to deal with disasters? (earthquake, power shutdown, fire...)

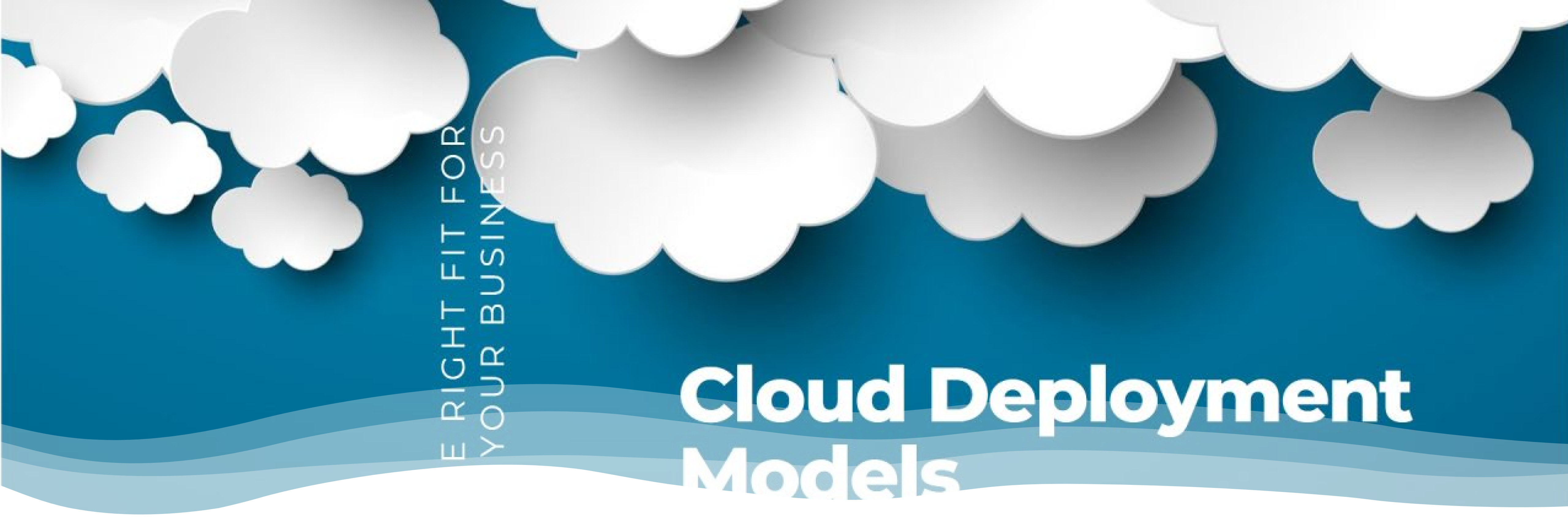
Can we externalize all this?

What is Cloud Computing

Cloud Computing is the delivery of computing services over the internet, enabling faster innovation, flexible resources, and economies of scale.

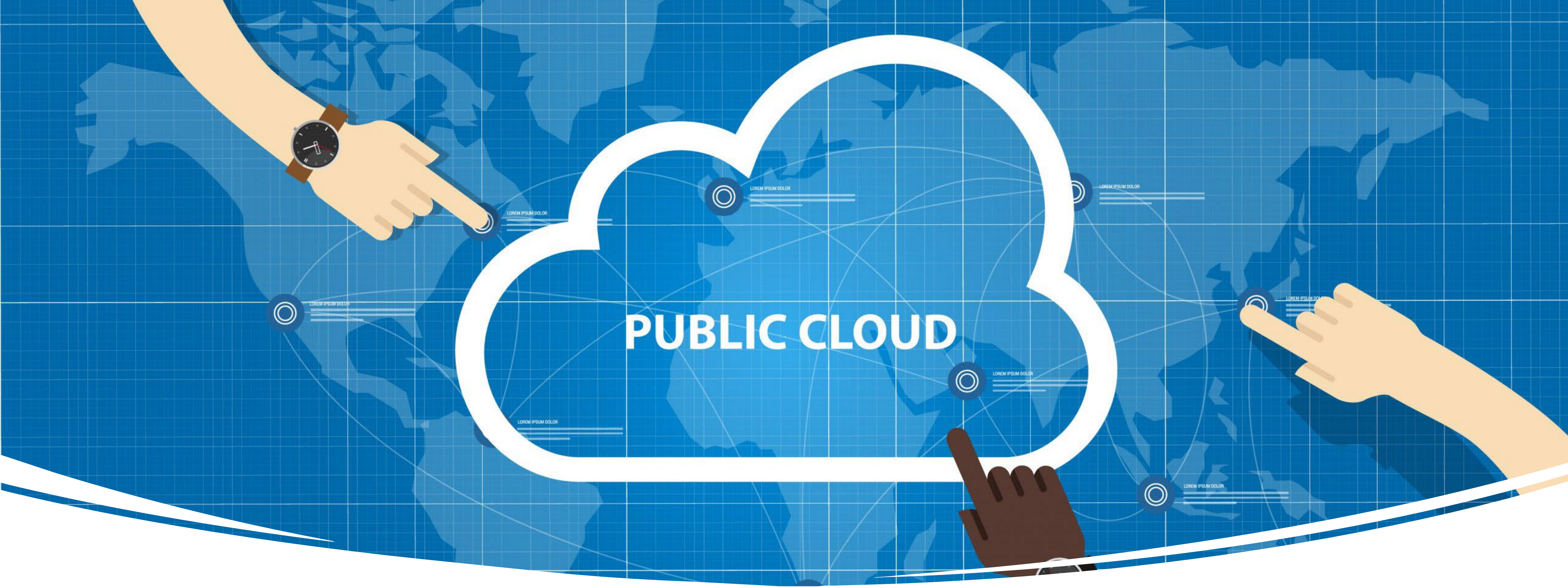
Cloud computing is the on-demand delivery of IT resources over the internet with pay-as-you-go pricing


v/s
ON-PREMISES
CLOUD



Cloud Deployment Models

When adopting cloud architecture, there are three different types of cloud deployment models that help deliver cloud computing services: public cloud, private cloud, and hybrid cloud.



Public Cloud

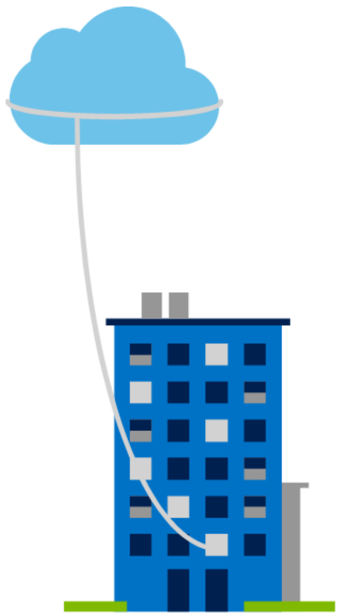
(Cloud-based Deployment)

- Public clouds deliver resources, such as compute, storage, network, develop-and-deploy environments, and applications over the internet.
- They are owned and run by third-party cloud service providers.

Google Cloud Platform
aws
ORACLE
CLOUD
Microsoft
Azure

Private Cloud

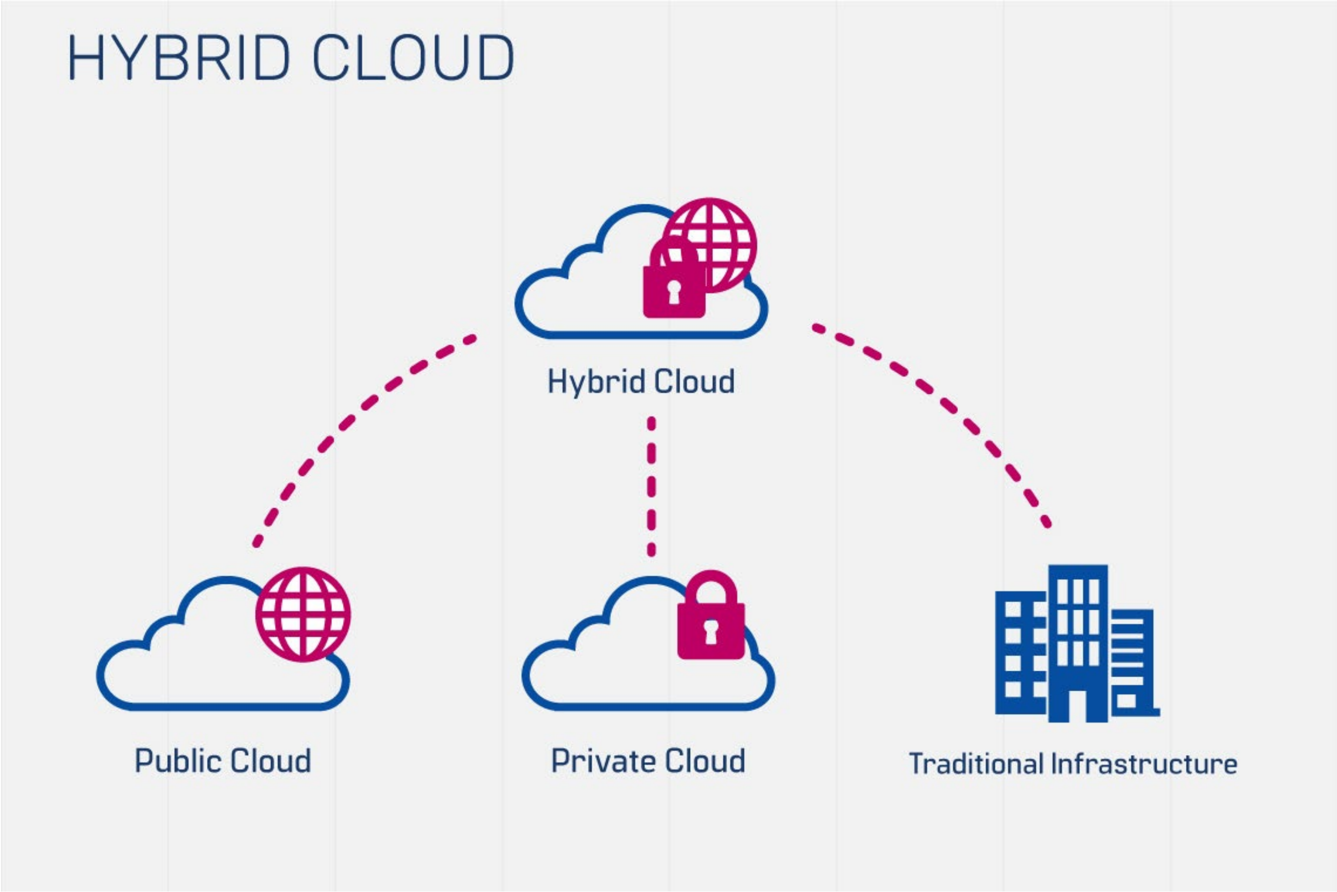
The deployment of resources on-premises, using virtualization and resource management tools, is sometimes called the private cloud.



Hybrid Cloud

Environments that mix at least one private computing environment (traditional IT infrastructure or private cloud, including edge) with one or more public clouds are called hybrid clouds.

Combines **Public** and **Private** clouds to allow applications to run in the most appropriate location.



HYBRID CLOUD

- Combination of both public and private cloud
- Shared security responsibility
- Helps maintain tighter controls over sensitive data and processes

PUBLIC CLOUD

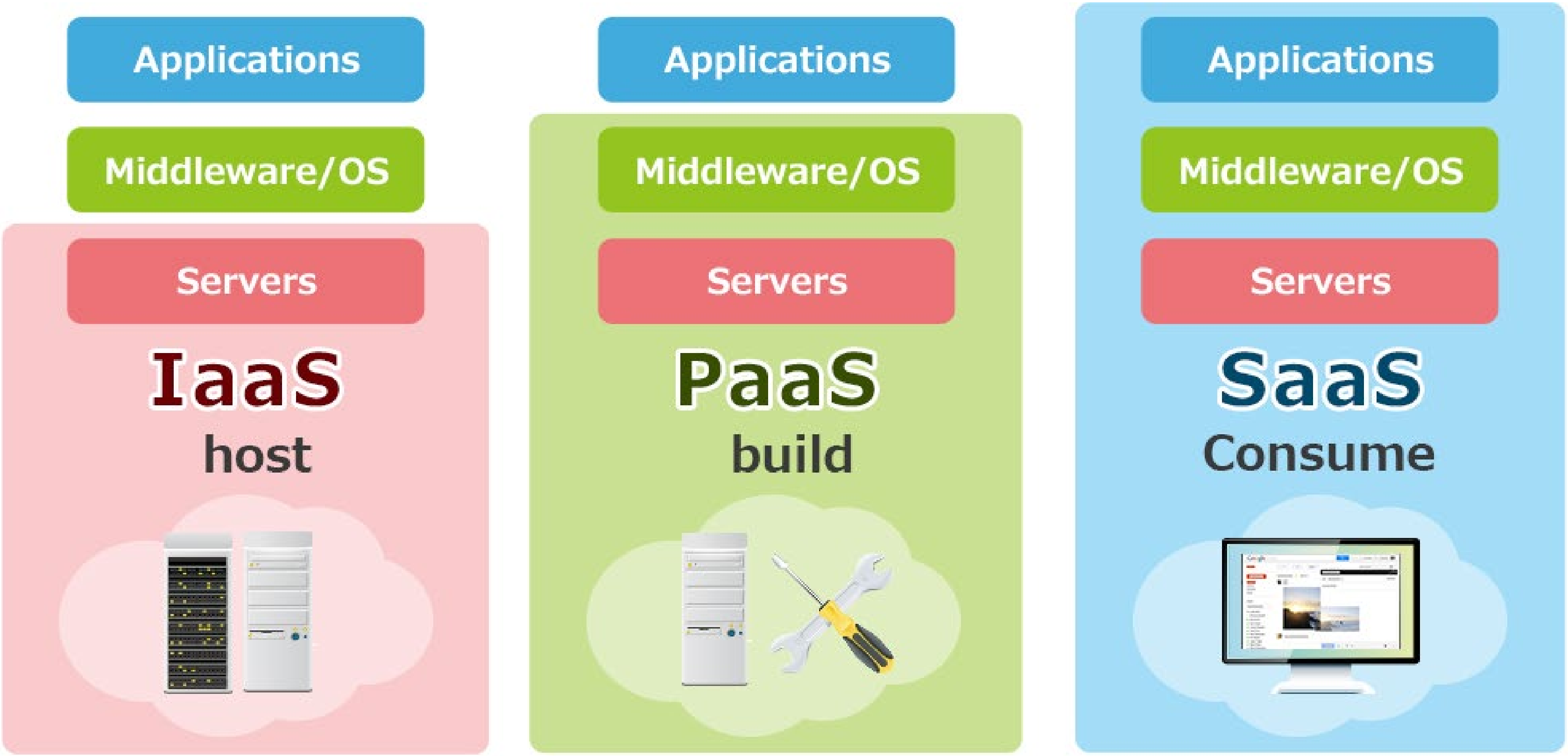
- Offered by third-party providers
- Available to anyone over the public internet
- Scales quickly and convenient

PRIVATE CLOUD

- Offered to select users over the internet or a private internal network
- Provides greater security controls
- Requires traditional datacenter staffing and maintenance

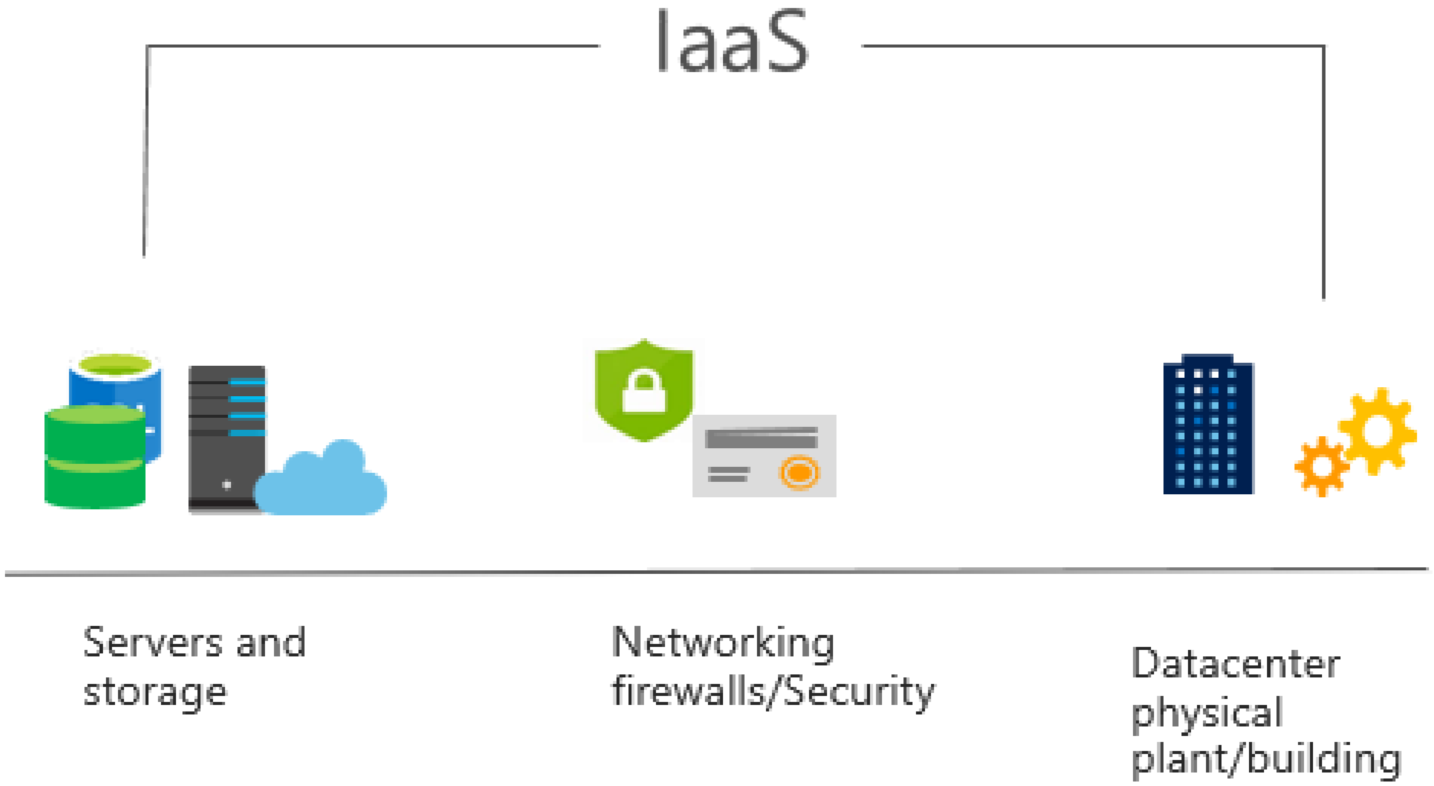
Cloud Computing Models

There are three main models for cloud computing. Each model represents a different part of the cloud computing stack.



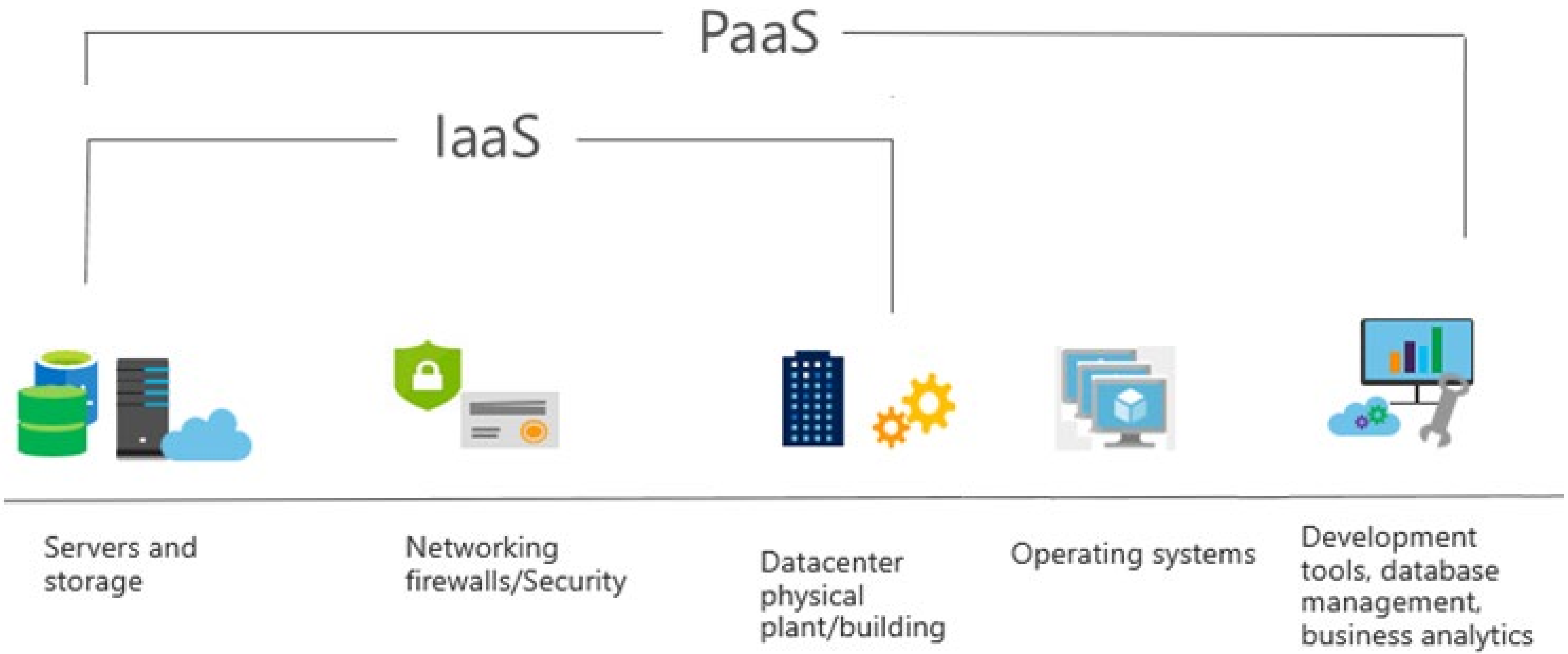
Infrastructure as a Service (IaaS)

- IaaS provides virtualized computing resources over the internet.
- It offers fundamental infrastructure like virtual machines, storage, and networking.



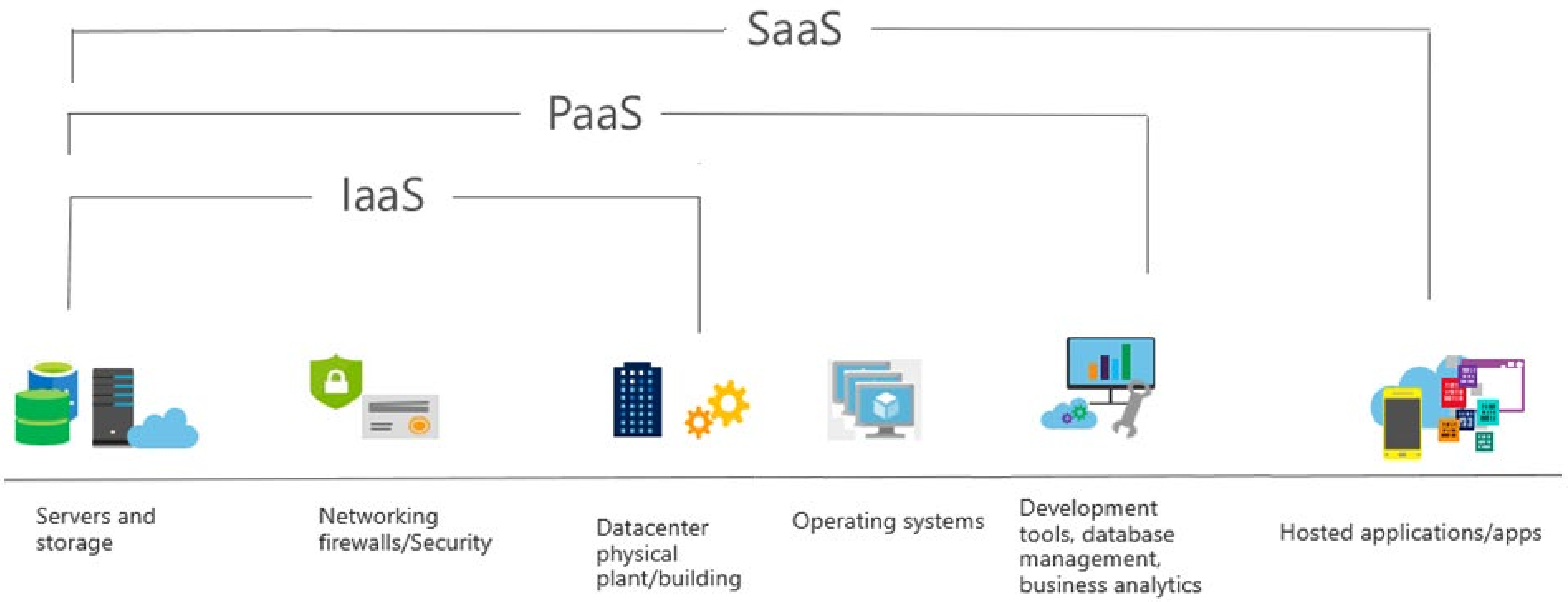
Platform as a Service (PaaS)

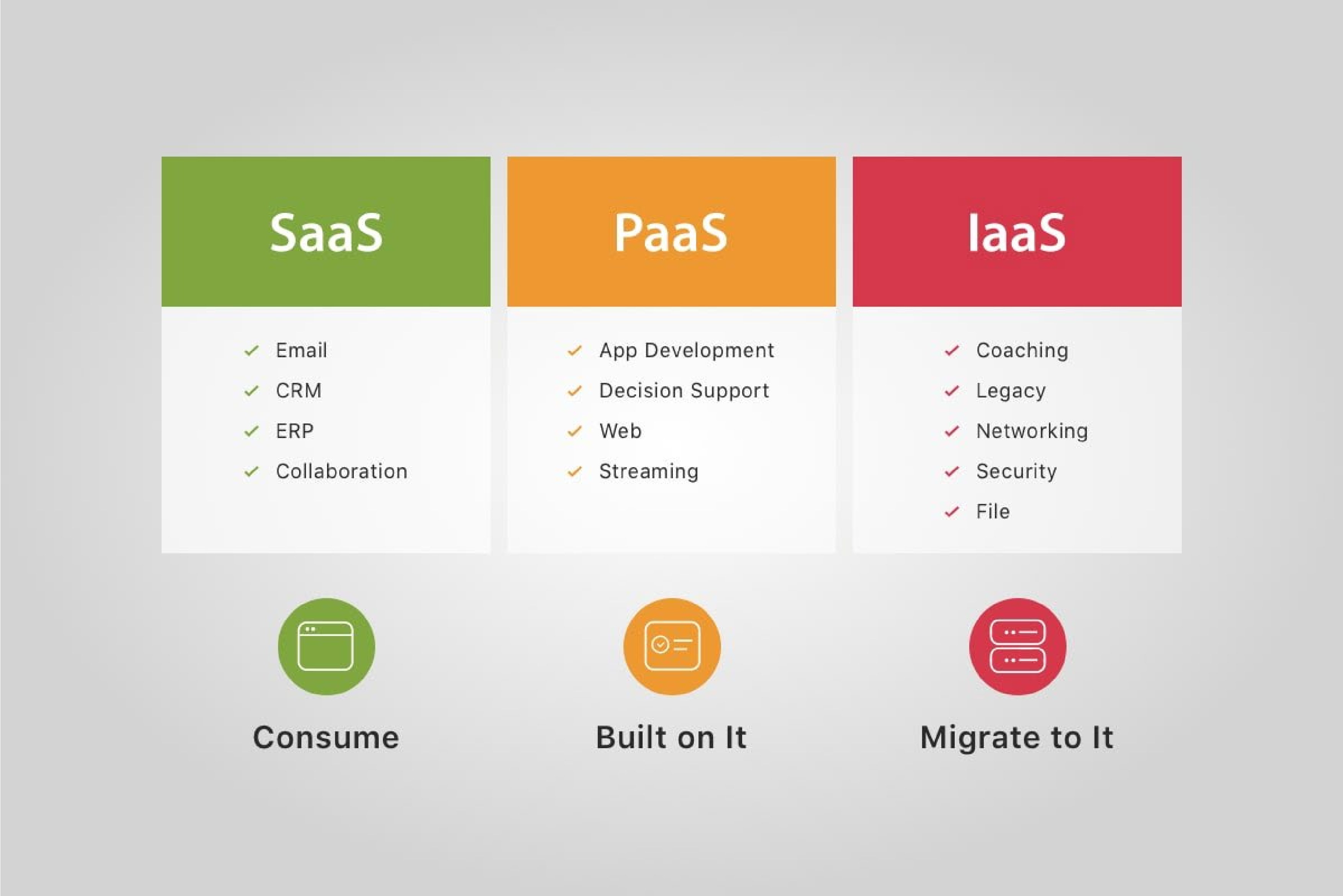
PaaS provides a platform allowing customers to develop, run, and manage applications without dealing with the underlying infrastructure.



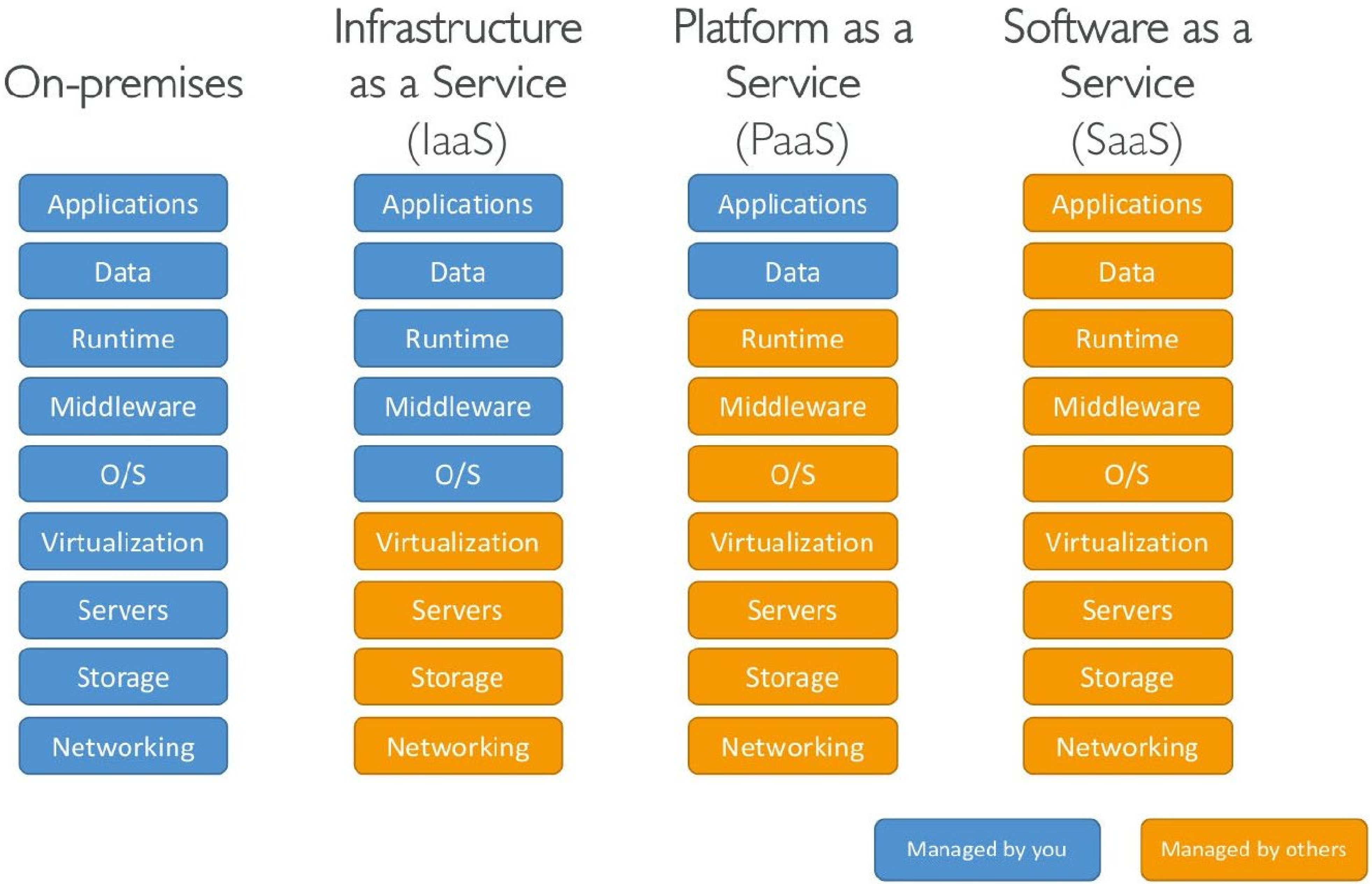
Software as a Service (SaaS)

- SaaS delivers software applications over the internet, on a subscription basis.
- Users access the software via a web browser.

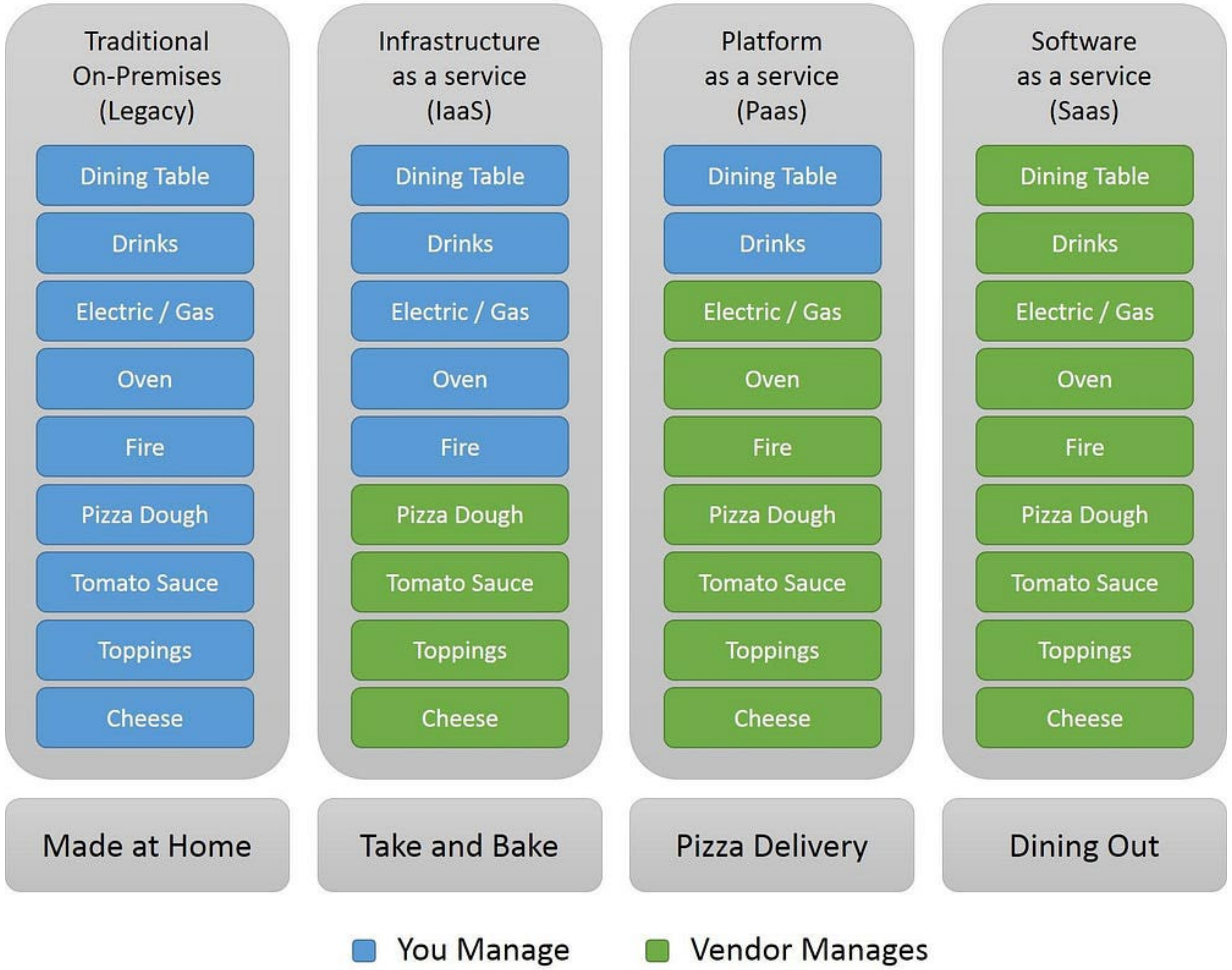



SaaS
Email
CRM
ERP
Collaboration
PaaS
App Development
Decision Support
Web
Streaming
IaaS
Coaching
Legacy
Networking
Security
File
Consume
Built on It
Migrate to It


IaaS
PaaS
SaaS
Amazon EC2
HEROKU
Gmail
阿里云
salesforce
PayPal
Azure
CLOUD
FOUNDRY
天億网络安全


On-premises
Infrastructure as a Service (IaaS)
Platform as a Service (PaaS)
Software as a Service (SaaS)
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Managed by you
Managed by others

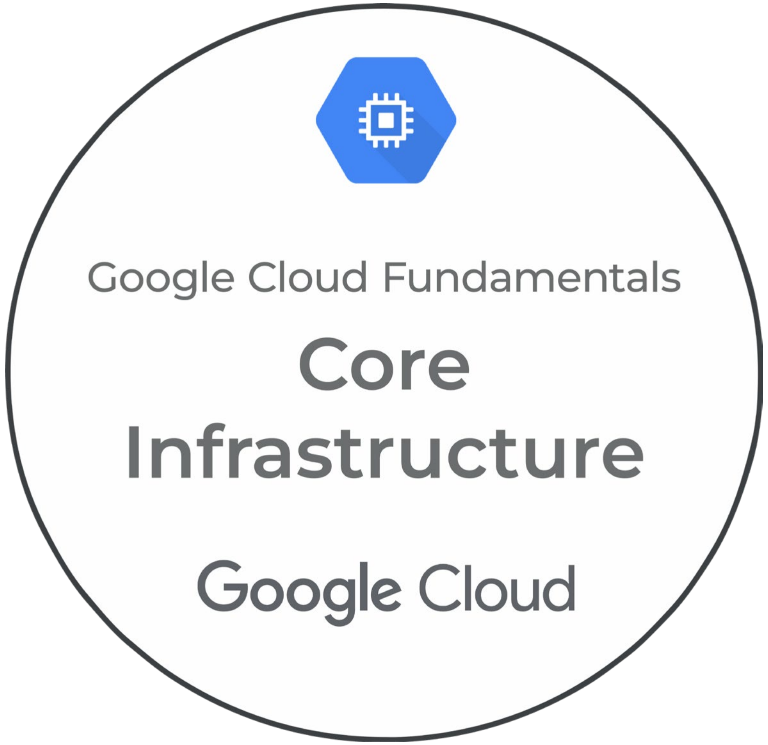
Pizza as a Service



Major Cloud Providers

The three leading cloud service providers are Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP.

How to start your cloud journey



Top Cloud Computing Careers

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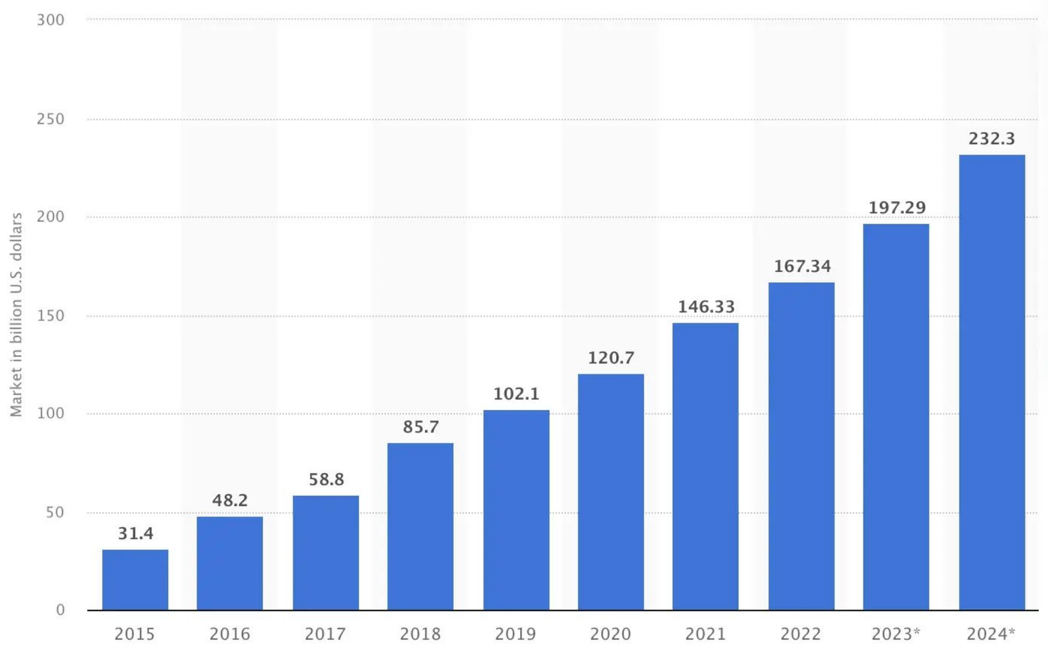
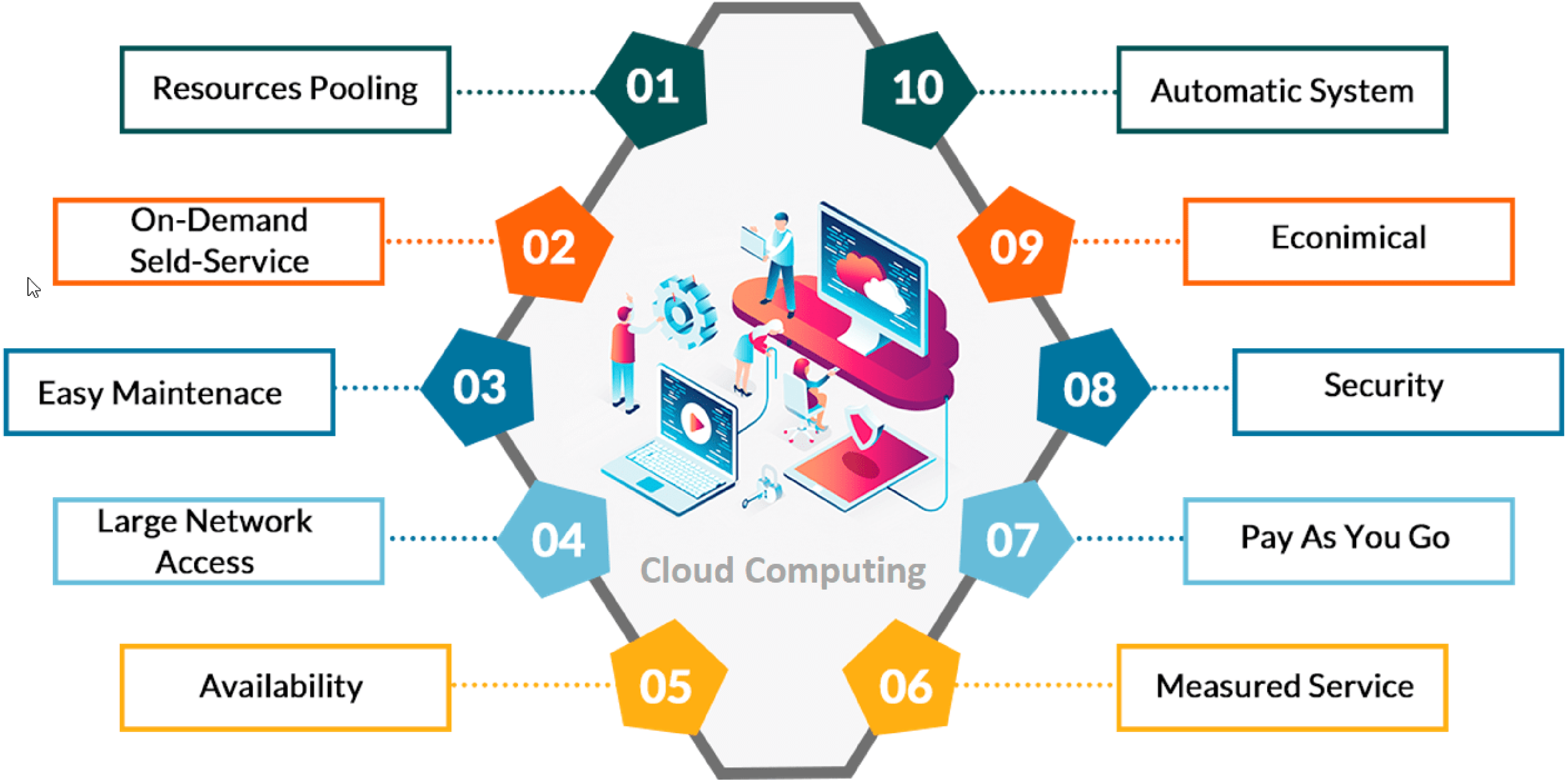
- Cloud Administrator
- Cloud Support Engineer
- Cloud Security Analyst
- Cloud Network Engineer
- Cloud Software Engineer
- Cloud Automation Engineer
- Cloud Engineer
- Cloud Consultant
- Cloud Data Scientist
- Cloud Architect
- Cloud Security Engineer
- Cloud DevOps Engineer


improving
As much as 80% of organizations are predicted to migrate towards the cloud by 2025
Forrester

Public Cloud Market Growth

The global public cloud computing market is on a steady rise and is projected to hit an estimated 232.3 billion U.S. dollars by 2024. This encompasses various services including business processes, platforms, infrastructure, software, management, security, and advertising. Witness the substantial growth depicted in the graph below, illustrating the ascending trajectory of the public cloud market over the years.




Resources Pooling
01
On-Demand Seld-Service
02
Easy Maintenace
03
Large Network Access
04
Availability
05
Cloud Computing
Measured Service
06
Pay As You Go
07
Security
08
Econimical
09
Automatic System
10

Cloud Computing Trends in 2025

- AI and ML
- Data Security
- Multi and Hybrid Cloud Deployment
- Low Code and No Code Cloud Solutions
- Edge computing
- IoT
- Kubernetes and Docker
- Serverless architecture/computing
- DevSecOps
- Disaster recovery and backup

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Let’s connect

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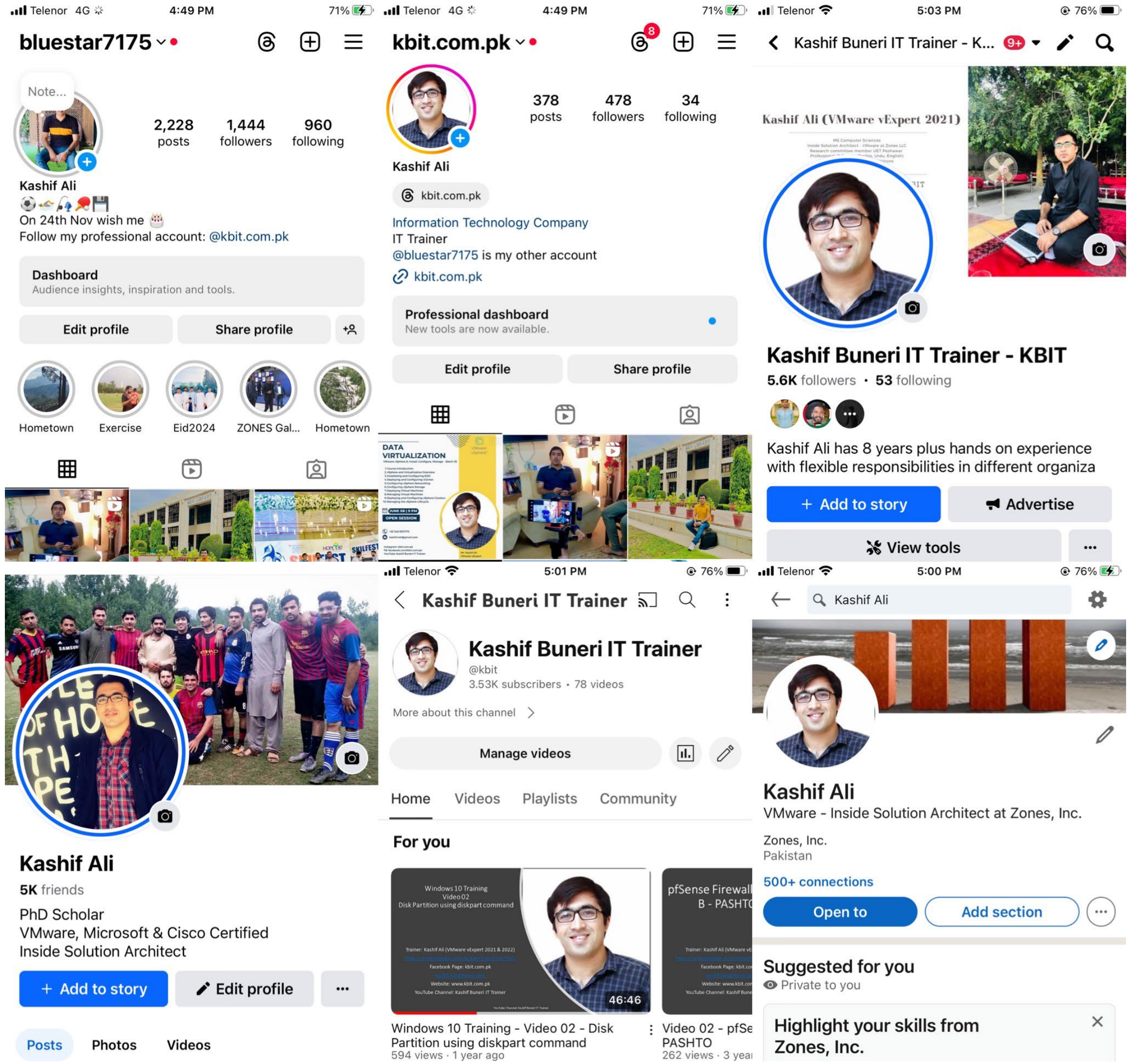
- YouTube: Kashif Buneri IT Trainer



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Kahoot!